AGENTIC AI PLATFORM ENGINEER · MULTI-AGENT RUNTIMES · POLICY-FIRST CONTROL PLANES

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Summary

Software engineer and architect with 20+ years of building and operating production systems, now focused entirely on agentic AI. Designed the layer that governs which tools and data six AI agents are allowed to use inside a HIPAA-regulated oncology prior authorization pipeline, together with the audit system that proves what each agent did, why it was permitted, and which model answered. Builds the whole path personally: agent orchestration, backend services and APIs, the database layer, and the React and command-line interfaces operators work in. Creator of Free2PA, which verifies signed agent instruction files before they reach a model. Six-time IBM Champion, winner of the IBM Watson Build Challenge North America, O'Reilly author of four books, and Co-Chair of the C2PA AI/ML Task Force.

Impact Beyond My Own Scope

Created capability that did not exist. Designed the capability abstraction layer now standing between six production agents and every backend they depend on, codified it as a project standard, and drift-audited the codebase against it — a foundation the team keeps building on.

Built leverage for other engineers. 13 internal Claude Code skills, including wiring generators that keep a growing agent codebase on its own standards, and PAM, a QA agent that acceptance-tests real user journeys and files field-level evidence.

Set direction beyond one company. Co-Chair of the C2PA AI/ML Task Force and contributor to SMPTE/ETC, turning provenance standards into implementations teams can actually adopt — then shipped Free2PA as the working reference.

Taught the field. Four O'Reilly books and the Radar article "AI's Opaque Box Is Actually a Supply Chain," plus mentorship that produced RadioHead, an award-winning agent now running at an NPR member station.

Core Skills

Agentic AI

Multi-Agent Orchestration · Custom Agent Harnesses & Runtimes · Ports-and-Adapters Capability Layers · Typed Capability Contracts & Registries · Model Context Protocol (MCP) · Function Calling & Structured Outputs · Human-Approval Gates & Recovery · Durable Session & Workflow State · RAG & Model Routing · Event-Driven Queue-Based Dispatch

Engineering

TypeScript, Node.js, React, Next.js · Python (intermediate) · PostgreSQL, Prisma, Schema Design & Migrations · Multi-Tenant APIs, OpenAPI · Azure AI Foundry, Azure OpenAI, Azure ML, FHIR · AWS Agent Deployment & Migration · Service Bus, Queue Workers, KEDA, SSE · OpenTelemetry, App Insights · CI/CD, Layered Testing, Release Gates

AgentOps & Governance

Hash-Chained Forensic Execution Traces · Token & Cost Accounting per Agent Step · Model Provenance Ledgers & Drift Detection · Least-Privilege Tool & Data Access · Prompt-Injection & Untrusted-Instruction Defenses · Claude Code Skill Authoring & Code Generators · Content Provenance (C2PA, SMPTE/ETC) · PHI-Safe Review Paths, RBAC / Managed Identity

Experience

Founder & Principal Engineer

Kilroy AI LLC

2026 – Present

- **Hidalga (client engagement)** — Architected the agent capability abstraction layer for a HIPAA-regulated six-agent oncology prior authorization pipeline: 16 domain-named capabilities over 24 swappable backends, so agents never bind to a vendor or model and backends change with zero agent code changes.
- Designed a typed capability outcome contract returning explicit unavailability rather than throwing, converting silent agent degradation into loud configuration errors, then codified the boundary as a project standard and drift-audited every agent against it.
- Converted four deterministic agents to LLM-backed agents on Azure AI Foundry for clinical extraction, document summarization, and payer narrative, while keeping every approval decision deterministic — the model reasons and writes, never renders the verdict.
- Built forensic agent observability: hash-chained tamper-evident execution traces with chain verification, liveness and stall detection, live SSE streaming, per-step token accounting, and model-attributed audit entries.
- Integrated an Azure ML prediction model behind a capability with a four-state trust contract, model-provenance ledger, and drift detection, so an unvalidated model declares itself rather than looking identical to a validated one.
- Authored 13 internal Claude Code skills including wiring code generators used as the anti-drift mechanism, and created PAM, a credential-free browser-driving QA agent that acceptance-tests user journeys and files field-level evidence.
- **RadioHead** — Built an autonomous broadcast transcription agent for an NPR member station, started on OpenClaw and migrated onto AWS for production reliability. Built with a student collaborator; won a student competition.
- **Free2PA** (free2pa.org) — Public provenance toolkit verifying signed agent control files before model context load: allow, deny, and quarantine primitives for a policy-first control plane, published as an open reference implementation.
- **Phyllis** (phyllis.bot) — Multi-tenant fulfillment API for autonomous commerce agents, with the consequential-action boundary designed in: agents prepare orders, humans approve anything that spends money or ships goods.

Founder & AI Engineer

NYX NoCode

2024 – Present

- Built and operate a platform where non-specialists describe an application in natural language and get a deployed React app — agent orchestration, model routing, RAG memory, code generation, and hosting, owned end to end.
- Serve public school customers in production, with the reliability, cost, accessibility, and safety obligations that implies.
- Ran hackathons and training programs that turned non-engineers into shipping builders.

Co-Chair, AI/ML Task Force

C2PA · contributor, SMPTE/ETC AI/ML Task Force

Current

- Co-chair cross-industry work on AI provenance, content authenticity, and standards adoption, then ship reference implementations so the standard becomes something teams can adopt.

Author

O'Reilly Media

2019 – 2024

- Four books: *Natural Language and Search* (2024), *Blockchain Tethered AI* (2023), *AI and the Law* (2021), *Blockchain as a Service* (2019), plus the O'Reilly Radar article "AI's Opaque Box Is Actually a Supply Chain."

CEO & Technical Lead

Kilroy Blockchain

2016 – 2025

- Led the team that built **RILEY**, winner of the IBM Watson Build Challenge North America (2017), an AI accessibility system for people who are blind or visually impaired.
- Directed engineering for blockchain-backed workflow and case management systems where auditability and operational control were core requirements.

Earlier Roles (Condensed)

- **CTO – Jamersan LLC** (2016) · **Principal Application Developer – CA Technologies** (2014–2015) · **Training Consultant – Magento Inc.** (2010–2014), founding member of Magento U · **Director of Online Marketing – Suarez Corp.** (2010–2012)

Recognition, Certifications & Education

IBM Watson Build Challenge Winner, North America (2017) · IBM Champion, six-time honoree (2020–2025)

AI Fluency for Students and Teaching the AI Fluency Framework — Anthropic (2025) · Venture Building — Builders + Backers (2025)

IBM Certifications: Watson Chatbot, RPA, Bluemix Essentials, Blockchain Essentials · FAA Private Pilot License

University of Arkansas, Sam M. Walton College of Business — Infrastructure & Cloud Computing (Spring 2026), 4.0 GPA; Quantum Computing (Fall 2026)

University of Arkansas — Studies in Music, 4.0 GPA (2023–Present) · **Hammel College** — Office Automation & Database Management (1981–1982), 4.0 GPA